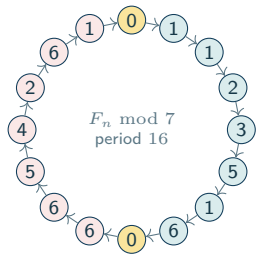
NUMBER THEORY LADDER 9 OF 9

Fibonacci modulo p and the golden ratio

Fibonacci repeats modulo every n, and whether a prime splits in the golden ratio field fixes the period.

1 / THE RULE AND THE PICTURE

Modulo 7 the sequence repeats every 16 terms, and its second half is the negative of its first



Reducing F_n modulo 7 gives $0, 1, 1, 2, 3, 5, 1, 6, 0, 6, 6, 5, 4, 2, 6, 1$ and then $0, 1, \ldots$ again, a cycle of period $\pi(7) = 16$, because a pair of consecutive terms determines the rest and there are finitely many pairs. The zero first reappears at $n = 8$ because $F_8 = 21 = 3 \cdot 7$, and then

$$F_{n+8} \equiv -F_n \pmod 7$$

so the sequence is the negative of itself half way round, and $16 = 2 \cdot 8$.

2 / THE CHARACTERISTIC EQUATION MODULO p

The roots of $x^2 - x - 1$ live in $\mathbb{F}_p$ or $\mathbb{F}_{p^2}$ according to the discriminant 5

Writing $F_n = (r_1^n - r_2^n)/(r_1 - r_2)$ with $r_{1,2} = \frac{1 \pm \sqrt{5}}{2}$ works modulo p whenever $\sqrt{5}$ exists.
If 5 is a square modulo p, that is $p \equiv \pm 1 \pmod 5$, the roots lie in $\mathbb{F}_p$ and $r^{p-1} = 1$, so the period divides $p - 1$. If 5 is not a square, $p \equiv \pm 2 \pmod 5$, the roots are conjugates with $r_1 r_2 = -1$, so $r_1^{p+1} = r_1 \cdot r_1^p = r_1 r_2 = -1$ and the period divides $2(p + 1)$. For $p = 5$ the double root 3 gives period 20.

3 / THE PERIODS / PRIMES UP TO 37

The period divides $p - 1$ for split primes and $2(p + 1)$ for inert ones

p	$p \bmod 5$	in $\mathbb{Z}[\varphi]$	period $\pi(p)$	first zero $\alpha(p)$	bound
2	2	inert	3	3	6
3	3	inert	8	4	8
5	0	ramified	20	5	20
7	2	inert	16	8	16
11	1	splits	10	10	10
13	3	inert	28	7	28
17	2	inert	36	9	36
19	4	splits	18	18	18
23	3	inert	48	24	48
29	4	splits	14	14	28
31	1	splits	30	30	30
37	2	inert	76	19	76

The first zero $\alpha(p)$, the rank of apparition, divides $p - \left(\frac{5}{p}\right)$ and the period is α, 2α or 4α: for example $p = 13$ has $\alpha = 7$ since $13 \mid F_7 = 13$, and $p = 11$ has $\alpha = 10$ since $F_{10} = 55 = 5 \cdot 11$.

4 / WHICH PRIMES DIVIDE WHICH FIBONACCI NUMBERS

Primes $\pm 1 \bmod 5$ split in the golden ratio field and the others stay prime

The ring $\mathbb{Z}[\varphi]$ has norm $a^2 + ab - b^2$, and a prime splits when $x^2 - x - 1$ has a root modulo p, which happens exactly when $\left(\frac{5}{p}\right) = 1$.
By reciprocity with $5 \equiv 1 \pmod 4$ this equals $\left(\frac{p}{5}\right)$, so the split primes below 100 are $11, 19, 29, 31, 41, 59, 61, 71, 79, 89$, and 5 is ramified.
Each prime divides some Fibonacci number: $17 \mid F_9 = 34$, $19 \mid F_{18} = 2584$, $23 \mid F_{24} = 46368$.

ASIDE / EVERY NUMBER DIVIDES ONE

Every integer divides infinitely many Fibonacci numbers, since 0 recurs in each period.

LINKS TO OTHER SHEETS

The Legendre symbol $\left(\frac{5}{p}\right)$ is in *The Legendre symbol and quadratic reciprocity*; the characteristic equation is in *The characteristic equation beneath five presentations*; the previous sheet is *Primes mod 3 and the Eisenstein integers*.

Convention. $\pi(p)$ is the Pisano period, $\alpha(p)$ the rank of apparition; $\varphi = \frac{1+\sqrt{5}}{2}$; $\mathbb{F}_p$ is the integers modulo p.